LEARNER GUIDE

Beyond the point: tenths, hundredths, thousandths

Place Value · Lesson 4. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

The columns carry on past the ones. Straight after the decimal point come **tenths**, then **hundredths**, then **thousandths**. Split one whole into ten equal parts and each is a tenth; split a tenth into ten and each is a hundredth; split a hundredth into ten and each is a thousandth. So 0.374 is 3 tenths, 7 hundredths and 4 thousandths.

What you are learning to notice

- Each column is a tenth of the column to its left, on both sides of the point.
- Zooming into a gap on a number line splits it into ten: each zoom adds one digit after the point.
- 0.07 is seven hundredths; seven tenths is 0.7. The column decides the value.
- Empty columns before the last digit need a 0: 4 tenths and 9 thousandths is 0.409.
- 0.5 and 0.50 are the same number: 5 tenths is 50 hundredths.

Moving through it, one step at a time

To write an amount as a decimal, draw the columns Ones . Tenths Hundredths Thousandths. Put each digit in its column, and a 0 in any empty column before the last digit. To find a decimal on a number line, use its tenths digit to choose the first gap, then zoom in and use the hundredths digit, then the thousandths digit.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Zooming in to 0.374

Where does 0.374 live on a number line from 0 to 1?

Zoom 1: 0.374 is between 0.3 and 0.4

Ones		Tenths	Hundredths	Thousandths
0	•	3		

Zoom 2: between 0.37 and 0.38

Ones		Tenths	Hundredths	Thousandths
0	•	3	7	

Zoom 3: found 0.374

Ones		Tenths	Hundredths	Thousandths
0	•	3	7	4

Each zoom splits one gap into ten, and adds one digit after the point.

1. The tenths digit is 3.	between 0.3 and 0.4
2. Zoom in. The hundredths digit is 7.	between 0.37 and 0.38
3. Zoom in again. The thousandths digit is 4.	0.374

0.374 = 3 tenths + 7 hundredths + 4 thousandths, which is 374 thousandths.

Example 2 — An empty column

Write 4 tenths and 9 thousandths as a decimal.

4 tenths and 9 thousandths

Ones		Tenths	Hundredths	Thousandths
0	•	4	0	9

The highlighted 0 holds the hundredths column, so the 9 stays in the thousandths.

1. Tenths digit	4
2. Hundredths: none, so hold the column with	0
3. Thousandths digit	9

0.409. Without the 0 it would be 0.49, which is 4 tenths and 9 hundredths.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (dividing by 10 (Lesson 2) and zero as a placeholder (Lesson 3)), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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